

Step 2

At $T = 1$ K and $B = 1$ T:

$$\mu_B B = 5.79 \times 10^{-5} \text{ eV} \approx 0.67 k_B T$$

$$\mu_N B = 3.15 \times 10^{-8} \text{ eV} \approx 3.7 \times 10^{-4} k_B T$$

$$\mu_B = 5.79 \cdot 10^{-5} \text{ eV/T}$$

$$\mu_N = 3.15 \cdot 10^{-8} \text{ eV/T}$$

$$\frac{\mu_B}{\mu_N} = \frac{m_p}{m_e} \approx 1836$$

$$k_B T = 8.62 \times 10^{-5} \text{ eV/K}$$

Electrons: feasible at $T = 1$ K

Nuclei: millikelvin temperatures required

Step 3

Choose an appropriate
cooling method